PHY201 Exam on chapter(s): 13, 14, 15

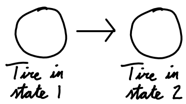
Name: *Sample*

These questions assess your understanding of the material from Chapters 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work. *full credit is awarded only for complete work. i.e. no points are awarded if complete work is not shown.*

1. Thomas inflates his bicycle tire to an absolute pressure of 0.49 MPa (a gauge pressure of about 56 psi) at a temperature of 7.8°C. What is the tire's gauge pressure after the temperature has risen to 37.2°C? Assume there are no leaks and the volume of the tire does not change.

ANSWER: 439.95 kPa

Derive an equation for P_{gauge}, P_g.



$$P_1V_1 = N_1k_BT_1 \qquad P_2V_2 = N_2k_BT_2$$

$$\frac{P_1\cancel{V_1}}{P_2\cancel{V_2}} = \frac{\cancel{N_1k_B}T_1}{\cancel{N_2k_B}T_2}$$

no leaks $\therefore N_1 = N_2$

Volume doesn't change $\therefore V_1 = V_2$

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$P_2 = P_1\left(\frac{T_2}{T_1}\right)$$

$$P_g + P_{atm} = P_1\left(\frac{T_2}{T_1}\right)$$

$$P_g = P_1\left(\frac{T_2}{T_1}\right) - P_{atm}$$

Get all in SI units

$$P_1 = 0.49\times10^6\ Pa$$

$$T_1 = 7.8 + 273.15 = 280.95\ K$$

$$T_2 = 37.2 + 273.15 = 310.35\ K$$

Solve

$$P_g = P_1\left(\frac{T_2}{T_1}\right) - P_{atm}$$

$$P_g = 0.49\times10^6\ Pa\left(\frac{310.35\ K}{280.95\ K}\right) - 101325\ Pa$$

$$P_g = 439951\ Pa$$

or 439.95 kPa

2. Linda puts a 0.77-kg aluminum pot on the stove and fills it with 0.46 liters of water. Both the pot and the water are initially at 3.1°C. How much heat is required to bring them both to 53.2°C? The specific heat value of water is 4186 J/kg°C and for aluminum it is 900 J/kg°C.

ANSWER: 131.19 kJ

Derive an equation for Q_{req}

$$Q_{req} = Q_{pot} + Q_{water}$$

$$Q_{req} = m_pc_p\Delta T_p + m_wc_w\Delta T_w$$

$$\because \Delta T_p = \Delta T_w = \Delta T$$

$$Q_{req} = (m_pc_p + m_wc_w)\Delta T$$

Get all in SI units

$$m_p = 0.77\ kg$$

$$c_p = 900\ J/kg°C = 900\ \frac{J}{kg\,K}$$

$$m_w = \rho_wV_w = 1000\frac{kg}{m^3}\left(0.46\ L\left|\frac{1\ m^3}{1000\ L}\right.\right)$$

$$c_w = 4186\frac{J}{kg°C} = 4186\frac{J}{kg\,K}$$

$$\Delta T = 53.2°C - 3.1°C = 50.1°C = 50.1\ K$$

Solve

$$Q_{req} = (m_pc_p + m_wc_w)\Delta T$$

$$Q_{req} = \left(0.77\ kg\cdot900\frac{J}{kg\,K} + 0.46\ kg\cdot4186\frac{J}{kg\,K}\right)50.1\ K$$

$$Q_{req} = 131190\ J$$

or 131.19 kJ

3. Calculate the work output of a Carnot engine operating between temperatures of 591 K and 417 K for 2550 J of heat transfer to the engine.

ANSWER: 750.8 J

Derive an equation for W

$$\eta = \frac{W}{Q_h}$$

$$\eta_c = 1 - \frac{T_c}{T_h}$$

$$\frac{W}{Q_h} = 1 - \frac{T_c}{T_h}$$

$$W = Q_h\left(1 - \frac{T_c}{T_h}\right)$$

Solve

$$W = Q_h\left(1 - \frac{T_c}{T_h}\right)$$

$$W = 2550\ J\left(1 - \frac{417\ K}{591\ K}\right)$$

$$W = 750.8\ J$$

4. Find the increase in entropy as 5.1 kg of ice, originally at 0.9°C, melts to form water at 0.9°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.

ANSWER: $6.22 \frac{kJ}{K}$

Derive an equation for ΔS

$$\Delta S = \frac{Q}{T}$$

$$Q = mL_f$$

$$\Delta S = \frac{mL_f}{T}$$

Solve

$$\Delta S = \frac{mL_f}{T}$$

$$\Delta S \approx \frac{(5.1\,kg)(334\frac{kJ}{kg})}{274.05\,K}$$

$$\Delta S = 6.22 \frac{kJ}{K}$$

5. In a particular year, the main span of San Francisco's Golden Gate Bridge was 1272-m long on the coldest day. The bridge was exposed to temperatures ranging from -6°C to 39°C. What is the change in its length between these temperatures? Assume that the bridge is made entirely of steel so that its coefficient of linear expansion is 12×10^{-6} 1/°C.

ANSWER: 0.687m

Derive an equation for ΔL

$$\Delta L = \alpha L \Delta T$$

Solve for ΔL

$$\Delta L = \alpha L \Delta T$$

$$\Delta L = (12 \times 10^{-6} \tfrac{1}{°C})(1272m)(39°C - (-6°C))$$

$$\Delta L = 0.687\,m$$

6. Sarah drives a 3200-kg truck and descends a 30-m-tall hill at a constant speed. What is the final temperature of 270 kg of brake material if its temperature is 6°C at the top of the hill? Assume the specific heat of the brake material is 850 J/kg°C. Assume the brakes do all the work to maintain the speed.

ANSWER: 10.1°C

Derive an equation for T_f

$$\Delta E_T = \left(\tfrac{1}{2}mv_f^2 + mgy_f\right) - \left(\tfrac{1}{2}mv_i^2 + mgy_0\right)$$

$$\Delta E_T = mg\Delta y \quad \because v_f = v_i$$

$$\Delta E_T = W_{nc} = W_{brakes}$$

$$\Delta U = 0 = Q - W_{brakes}$$

$$W_{nc} = Q$$

$$Q = m_b c \Delta T$$

$$mg\Delta y = m_b c (T_f - T_i)$$

$$T_f = \frac{mg\Delta y}{m_b c} + T_i$$

Solve for T_f

$$T_f = \frac{mg\Delta y}{m_b c} + T_i$$

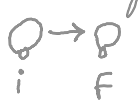
$$T_f = \frac{(3200kg)(9.8\frac{m}{s^2})(30m)}{(270kg)(850\frac{J}{kg°C})} + 6°C$$

$$T_f = 10.1°C$$

7. Emma is a design engineer for a lightbulb company. She manufactures a gas-filled incandescent light bulb so that the gas inside the bulb is at atmospheric pressure when the bulb has a temperature of 5°C. What is the pressure in the light bulb when it rises to 238°C? The coefficient of volume expansion for the glass light bulb is 32×10^{-6} 1/°C.

ANSWER: 184 kPa

Derive an equation for P_f



$\Delta N = 0$

no change in amount of gas in bulb

ie. no leaks

$P_iV_i = N_iK_BT_i \therefore N_i = \dfrac{P_iV_i}{K_BT_i}$

$P_FV_F = N_FK_BT_F \therefore N_F = \dfrac{P_FV_F}{K_BT_F}$

$\because \Delta N = 0$

$\dfrac{P_FV_F}{K_BT_F} - \dfrac{P_iV_i}{K_BT_i} = 0$

$\dfrac{P_FV_F}{T_F} = \dfrac{P_iV_i}{T_i}$

$\Delta V = V_F - V_i = \beta V_i \Delta T$

$\therefore V_F = V_i(1+\beta(T_F - T_i))$

$P_F = \dfrac{T_F}{T_i}\dfrac{V_i}{V_F}P_i$

$P_F = \dfrac{T_F}{T_i}\dfrac{V_i}{V_i(1+\beta(T_F-T_i))}P_i$

$P_F = \dfrac{T_F}{T_i}\dfrac{P_i}{1+\beta(T_F-T_i)}$

Solve

$P_F = \dfrac{T_F}{T_i}\dfrac{P_i}{1+\beta(T_F-T_i)}$

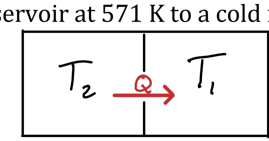
$P_F = \left(\dfrac{511.15K}{278.15K}\right)\dfrac{101325\ Pa}{\left(1+32\times10^{-6}\frac{1}{°C}(233°C)\right)}$

$P_F = 184\ kPa$

8. Spontaneous heat transfer from hot to cold is an irreversible process. Calculate the total change in entropy if 6120 J of heat transfer occurs from a hot reservoir at 571 K to a cold reservoir at 254 K, assuming that there is no temperature change in either reservoir.

ANSWER: $13.38\ \frac{J}{K}$

Derive an equation for ΔS_T



$T_2 > T_1$

Room 2 $T_2 = 571K$ "hot reservoir"

Room 1 $T_1 = 254K$ "cold reservoir"

$\Delta S_{total} = \Delta S_{room\ 2} + \Delta S_{room\ 1}$

$\Delta S_2 = \dfrac{-Q}{T_2} \qquad \Delta S_1 = \dfrac{Q}{T_1}$

$\therefore \Delta S_{total} = \dfrac{-Q}{T_2} + \dfrac{Q}{T_1}$

$\Delta S_{total} = Q\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$

Solve for ΔS_{total}

$\Delta S_{total} = Q\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$

$\Delta S_{total} = 6120J\left(\dfrac{1}{254k} - \dfrac{1}{571\,K}\right)$

$\Delta S_{total} = 13.38\ \dfrac{J}{K}$

9. Joseph is producing heat at a rate of about 244 W while doing light exercise. At what rate must water evaporate from his body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C.

ANSWER: 6.02 g/min

Derive an equation for $\frac{m}{t}$ or $\dot{m}$, the rate of loss of sweat mass

$\Delta U = Q - W_{Joseph}$

$\dfrac{\Delta U}{t} = \dfrac{Q}{t} - \dfrac{W_J}{t}$

$\Delta\dot{U} = \dot{Q} - \dot{W}_J$

$\boxed{\begin{array}{l} \Delta\dot{U} = 0\ W \because \text{no internal energy accumulates in Joseph. His body gets rid of all produced energy} \\ \dot{Q} = \dfrac{Q}{t} = \dfrac{mL_v}{t} = \dot{m}L_v \\ \dot{W}_J = \dfrac{W_{Joseph}}{t} = -244\ W \because \text{work is done on him to increase his energy } U \end{array}}$

$0 = \dot{Q} - \dot{W}_J$

$\dot{Q} = \dot{W}_J$

$\dot{m}L_v = \dot{W}_J$

$\dot{m} = \dfrac{\dot{W}_J}{L_v}$

Solve for $\dot{m}$

$\dot{m} = \dfrac{\dot{W}_J}{L_v}$

$\dot{m} = \dfrac{-244\frac{J}{s}\left(\frac{60\,s}{1\,min}\right)}{2430000\frac{J}{kg}\left(\frac{1\,kg}{1000\,g}\right)}$

$\dot{m} = -6.02\ g/min$

mass loss indicated by $\dot{m} < 0$

10. Samantha's 50-L (13.2-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 6.1°C. After both the tank and the gasoline warm to 35.7°C, what is the pressure in the gasoline tank if the gasoline cannot spill out? The coefficient of volume expansion for steel is 35×10^{-6} 1/°C and for gasoline it is 950×10^{-6} 1/°C. Assume the bulk modulus B for gasoline is 1.00×10^9 N/m^2.

ANSWER: 27.1 MPa

Derive an equation for P

$\Delta V_s = \beta_s V_1 \Delta T$

$\Delta V_g = \beta_g V_1 \Delta T$

these equations describe the thermal expansion of the steel & the gasoline

$V^* = \Delta V_g - \Delta V_s$

Notice that the gas expands more than the tank does. V^* is the volume of gas that is not accommodated by the tank.

$V^* = \beta_g V_1 \Delta T - \beta_s V_1 \Delta T$

$V^* = V_1 \Delta T (\beta_g - \beta_s)$

$\Delta V = \frac{1}{B} \frac{F}{A} V_0$

$V^* = \frac{1}{B} P V_1$

$P = B \frac{V^*}{V_1}$

$P = B \Delta T (\beta_g - \beta_s)$

this equation describes the pressure exerted on the steel tank as the excess gasoline forces it to expand further by V^*

Solve for P

$$P = B \Delta T (\beta_g - \beta_s)$$

$$P = 1 \times 10^9 \text{ Pa} (29.6°C)\left((950 - 35) \times 10^{-6} \tfrac{1}{°C}\right)$$

$$P = 27.1 \text{ MPa}$$

11. What is the average kinetic energy of a gas molecule at 94°C?

ANSWER: 7.6×10^{-21} J

$$\overline{KE} = \frac{3}{2} k_B T$$

$$\overline{KE} = \frac{3}{2}\left(1.38 \times 10^{-23} \tfrac{J}{K}\right)(94 + 273.15)K$$

$$\overline{KE} = 7.6 \times 10^{-21} J$$

12. Tyler's 51-L (13.5-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 5.4°C. How much gasoline has leaked out due to thermal expansion by the time both the tank and the gasoline have warmed to 29.1°C? The coefficient of volume expansion for steel is 35×10^{-6} 1/°C and for gasoline it is 950×10^{-6} 1/°C.

ANSWER: 1.106 L

Derive an equation for V_{leak} or V^*

$\Delta V_s = \beta_s V_1 \Delta T$

$\Delta V_g = \beta_g V_1 \Delta T$

these equations describe the thermal expansion of the steel & the gasoline

$V^* = \Delta V_g - \Delta V_s$

Notice that the gas expands more than the tank does. V^* is the volume of gas that is not accommodated by the tank.

$V^* = \beta_g V_1 \Delta T - \beta_s V_1 \Delta T$

$V^* = V_1 \Delta T (\beta_g - \beta_s)$

Solve for V^*

$$V^* = V_1 \Delta T (\beta_g - \beta_s)$$

$$V^* = (51 \text{ L})(29.1°C - 5.4°C)\left((950 - 35) \times 10^{-6} \tfrac{1}{°C}\right)$$

$$V^* = 1.106 \text{ L}$$